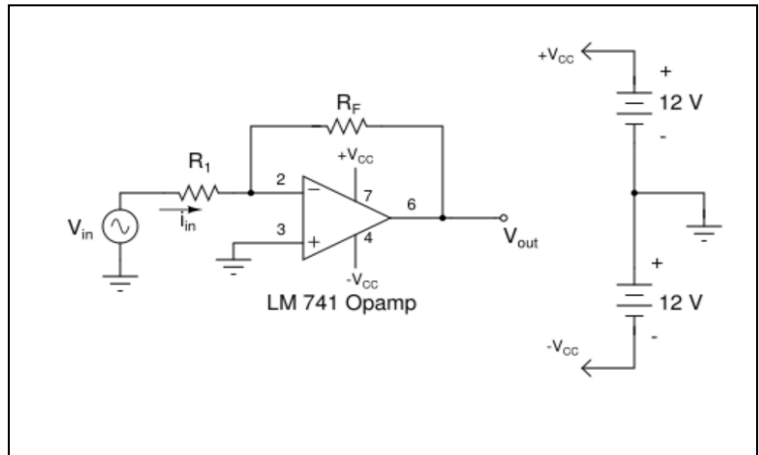


## Pre-lab Quiz for EE Expt 2 / Sample Set

Duration: 10 min. Marks:  $5 \times 1$  mark = 5 marks. No partial marks for any of the questions. Do your rough work in this paper itself. No extra sheets will be given. No clarifications/explanations will be given.

- 1) The inverting amplifier circuit of Expt 2 is shown in the figure, Given:  $v_{in}(t) = 0.5 \sin(\omega t + \theta)$  V of frequency 1 kHz,  $R_1 = 4 \text{ k}\Omega$ ,  $R_F = 1 \text{ k}\Omega$ . Mark **all the correct options** with regard to this circuit.



- A) The voltage gain is  $-4$ .  
~~B) The voltage gain is  $-0.25$ .~~  
 C)  $V_{in}$  is a differential signal, and  $V_{out}$  is a single-ended signal.  
~~D)  $V_{in}$  is a single-ended signal, and  $V_{out}$  is also a single-ended signal.~~  
~~E) The input resistance ( $V_{in}/I_{in}$ ) is  $4 \text{ k}\Omega$ .~~  
 F) The input resistance ( $V_{in}/I_{in}$ ) is  $1 \text{ k}\Omega$ .  
~~G) The circuit qualifies as an amplifier because it can provide a power gain larger than unity.~~  
 H) The circuit qualifies as an amplifier because the output voltage amplitude is larger than the input voltage amplitude.

Answer(s): \_\_\_\_\_

- 2) In the circuit shown above,  $R_1 = 2 \text{ k}\Omega$ ,  $R_F = 20 \text{ k}\Omega$ . Assume that the  $v_{out}$  limits are  $+V_{cc}$  and  $-V_{cc}$ . Mark **all the correct options** with regard to this circuit.
- A) The input  $v_{in}(t)$  is a triangular wave going from  $-2 \text{ V}$  to  $+2 \text{ V}$ . The output  $v_{out}(t)$  will also be a triangular wave.  
 B) The input  $v_{in}(t)$  is a square wave with the maximum and minimum levels of  $+2 \text{ V}$  and  $-2 \text{ V}$ . The output  $v_{out}(t)$  will be a square wave with the maximum and minimum levels of  $+20 \text{ V}$  and  $-20 \text{ V}$ .  
~~C) The input  $v_{in}(t)$  is a square wave with the maximum and minimum levels of  $+2 \text{ V}$  and  $-2 \text{ V}$ . The output  $v_{out}(t)$  will be a square wave with the maximum and minimum levels  $+12 \text{ V}$  and  $-12 \text{ V}$ .~~  
~~D) If the connections to the op amp input terminals are interchanged, the circuit will not work as a linear amplifier.~~  
 E) If the connections to the op amp input terminals are interchanged, the circuit will work as a non-inverting amplifier.

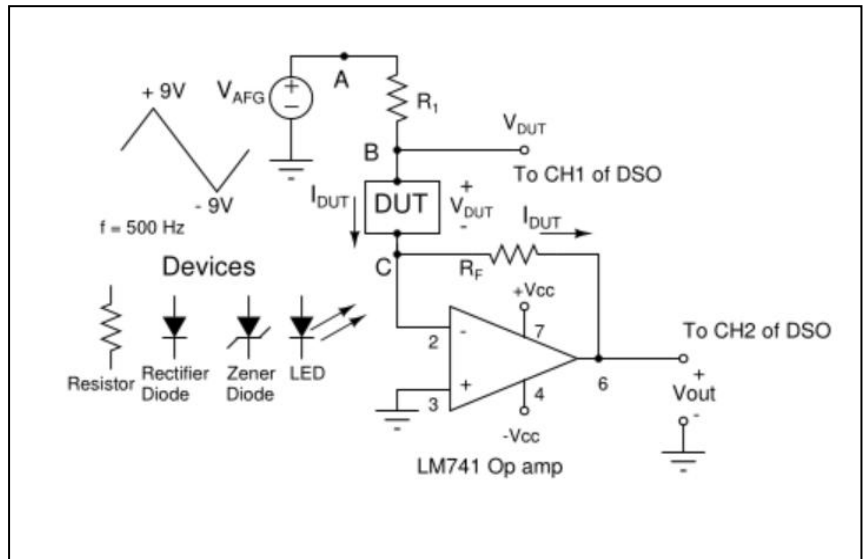
Answer(s): \_\_\_\_\_

3&4) The current-to-voltage converter circuit of Expt 2 is shown in the figure. Given:  $R_1 = 10\text{ k}\Omega$ ,  $R_F = 20\text{ k}\Omega$ ,  $+V_{cc} = +12\text{ V}$ ,  $-V_{cc} = -12\text{ V}$ .

A rectifier diode is connected as the device-under-test (DUT) with its cathode at terminal B and anode at terminal C. The diode I-V characteristic has a forward voltage drop of  $0.6\text{ V}$ , and negligible reverse saturation current. Calculate the current through the device ( $I_{DUT}$ ) and the voltage across it ( $V_{DUT}$ ) at the instant when  $V_{AFG} = +5.6\text{ V}$ .

$$I_{DUT} = \underline{\quad 0.5 \quad} \text{ mA}$$

$$V_{DUT} = \underline{\quad 0.6 \quad} \text{ V}$$



5) In the circuit shown above,  $R_1 = 10\text{ k}\Omega$ ,  $R_F = 20\text{ k}\Omega$ ,  $+V_{cc} = +12\text{ V}$ ,  $-V_{cc} = -12\text{ V}$ . A resistor of  $2\text{ k}\Omega$  is connected across the nodes B and C as the device-under-test (DUT). Calculate output voltage ( $V_{out}$ ) at the instant when  $V_{AFG} = -3\text{ V}$ .

$$V_{out} = \underline{\quad +5.0 \quad} \text{ V}$$

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